Problem statement 1: Teacher vs Student Identity

- Teacher → role();
- Student → role()
- Person → inherit Both.

Problem Statement 2: ~~Dat~~ Salary system.

- full Time → salary()
- Part Time → salary()
- Employee → Both. inherit.

Problem statement 3: Device Control system.

- Camera → function start()
- Music Player → Start()
- smartDevice → inherit Both.

Problem

Compiler doesn't know.
use A::show() or B::show()

Solution (Scope Resolution operator (::))
We tell compiler which one to use.

Fixed code

```
int main() {
    C obj;

    obj.A::show();  // A
    obj.B::show();  // B
}.
```

Another Example with (variables)

```
class A {
    public:
        int x = 10;
};

class B {
    public:
        int x = 20;
};

class C: public A, public B {};

int main() {
    C obj;

    // cout obj.x;  X (Ambigous)
    cout << obj.A::x << endl;  // 10
    cout << obj.B::x << endl;  // 20
}
```

Problem statement 1: Smart Device System

Classes
- camera → take Photo()
- music Player → PlayMusic()
- smartphone → inherit both.

output:
- capture Photo
- Playmusic

Problem statement 2: Hybrid Vehicle System.

Classes
- PetrolEngine → petrol info.
- ElectricEngine → battery info
- Hybrid car → inherit both.

display:
- fuel type
- battery level.

Problem statement 3: Student Employee System.

Classes
- student → marks
- Employee → salary
- intern → inherits both.

display
- marks
- salary
- internship status.

Day 38 C Ambiguity resolution in inheritance)

Ambiguity happens when compiler get confused because same function or variable exist in multiple parent class

When does it happen?
In multiple inheritance

Parent 1 → Same function ← Parent 2
↓
Child

Example